

Personalized Latent State Discovery and Temporal Modeling of Longitudinal Wearable Physiology: A Leakage-Controlled Cross-Domain Evaluation Framework

S. Lokesh

*Department of Information Science and Engineering
Nitte Meenakshi Institute of Technology
Bengaluru, India
mail2lokesh022@gmail.com*

Nigam L. Raj

*Department of Information Science and Engineering
Nitte Meenakshi Institute of Technology
Bengaluru, India
nigamraj711@gmail.com*

Prajwal R.

*Department of Information Science and Engineering
Nitte Meenakshi Institute of Technology
Bengaluru, India
prajwalgowda81234@gmail.com*

Sachin S. K.

*Department of Information Science and Engineering
Nitte Meenakshi Institute of Technology
Bengaluru, India
sachinkagale11@gmail.com*

Lakshmi M.

*Department of Information Science and Engineering
Nitte Meenakshi Institute of Technology
Bengaluru, India
lakshmi.m@nmit.ac.in*

Abstract—Continuous physiological monitoring via consumer wearable devices offers unique opportunities for longitudinal personal health tracking. However, modeling wearable time series is severely complicated by high inter-individual baseline variation, non-stationary daily dynamics, missing data, and an absence of continuous clinical labels. In this paper, we present a leakage-controlled machine learning framework for unsupervised latent health state discovery and dynamic sequence decoding across synthetic and real-world wearable cohorts. By transforming multi-modal signals into personalized 7-day rolling baseline deviations, Gaussian Mixture Models (GMM) discover three macro-level physiological states, followed by Hidden Markov Model (HMM) sequence decoding. Evaluating across 69 real Fitbit users (4,159 daily timelines) and 300 synthetic users (55,200 timelines), zero-shot cross-domain transfer (synthetic to real) achieves a Silhouette score of 0.2117 vs. 0.2168 for real-only baseline. Applying Nadeau-Bengio resampled variance corrections and Holm-Bonferroni multiple-comparison adjustments across ratio sweeps ($p_{\text{adj}} < 0.004$), we demonstrate that synthetic data enables effective zero-shot model initialization without touching real user biometrics during training, while synthetic data augmentation provides negligible benefit when real-world training data is already abundant.

Index Terms—*Wearable Computing, Unsupervised Learning, Gaussian Mixture Models, Hidden Markov Models, Cross-Domain Transfer.*

I. INTRODUCTION

Continuous physiological monitoring via consumer wearable devices offers unique opportunities for longitudinal personal health tracking. However, modeling wearable time series is severely complicated by high inter-individual baseline variation, non-stationary daily dynamics, missing data, and an absence of continuous clinical labels. Consumer wearable devices such as smartwatches and fitness trackers continuously capture multi-modal physiological streams including resting heart rate, daily physical activity, calorie expenditure, and sleep duration. These longitudinal data streams provide unprecedented granularity for personal health tracking. However, translating raw wearable metrics into actionable physiological insights presents major

technical hurdles. First, raw sensor streams exhibit high inter-individual baseline variation; a resting heart rate of 70 BPM may represent elevated physiological stress for an athletic individual but normal baseline for a sedentary user. Second, real-world wearable datasets suffer from severe sensor missingness, non-stationary routine shifts, and a complete absence of continuous clinical diagnostic labels.

To address these challenges without relying on supervised clinical targets, unsupervised learning approaches such as Gaussian Mixture Models (GMM) and Hidden Markov Models (HMM) have emerged as powerful paradigms for latent state discovery. Furthermore, privacy concerns and dataset scarcity make synthetic data generation an attractive alternative for model pre-training. However, whether latent state structures learned

from synthetic wearable distributions generalize zero-shot to real-world human physiological timelines remains an open research question.

In this work, we present an end-to-end unsupervised framework for personalized physiological state discovery and cross-domain transfer evaluation. Our core contributions are:

1. A causal, leakage-controlled preprocessing framework that converts multi-modal wearable time series into personalized 7-day rolling z-deviations and a harmonized composite severity score.
2. A comparative four-experiment cross-domain evaluation framework (Synthetic-to-Synthetic, Real-to-Real, Synthetic-to-Real Zero-Shot, and Synthetic+Real Ratio Sweeps), establishing 50-shuffle permutation null baselines for all clustering evaluations.
3. Rigorous cross-domain statistical evaluation utilizing Nadeau-Bengio resampled variance corrections and Holm-Bonferroni multiple-comparison adjustments to benchmark zero-shot transferability and ratio augmentation.

We emphasize that the discovered clusters—interpretively labeled *Recovery*, *Baseline*, and *Strain* by sorting GMM components by median harmonized severity—represent heuristic data-driven proxy labels for exploratory wellness tracking, not clinical diagnoses.

End-to-End Leakage-Controlled Latent State Discovery Pipeline



Fig. 1. End-to-end leakage-controlled latent state discovery pipeline operating across data ingestion, causal preprocessing, GMM density estimation, HMM sequence decoding, and Streamlit application display.

II. RELATED WORK

Longitudinal physiological state discovery from wearable sensors builds upon time-series feature engineering and probabilistic sequence modeling. Prior work has demonstrated that raw physiological metrics fail to generalize across diverse user cohorts due to static demographic offsets. To mitigate baseline heterogeneity, recent studies advocate per-subject z-score normalization over rolling temporal windows.

Gaussian Mixture Models (GMM) provide a probabilistic framework for soft clustering, estimating continuous posterior probabilities $P(\text{State}_k \mid \mathbf{x}_t)$ that reflect state assignment uncertainty. Sequential dynamics are subsequently captured via Hidden Markov Models (HMM), where Viterbi decoding infers optimal latent state sequences over time. Clustering quality in unsupervised domain adaptation is typically evaluated using internal validation metrics such as the Silhouette coefficient, Davies-Bouldin index (DBI), and Calinski-Harabasz (CH) index. However, standard statistical comparisons across resampled cross-validation folds often suffer from variance underestimation; Nadeau and Bengio formulated a corrected resampled t-test to account for non-independence across resamples.

III. RESEARCH QUESTIONS AND CONTRIBUTIONS

Our experimental evaluation is governed by three locked Research Questions (RQs):

- **RQ1 (In-Domain Baseline Separation):** Can personalized z-deviation transformations establish statistically meaningful cluster separation ($\text{Silhouette} > \text{Null Floor}$) over raw physiological signals within synthetic and real domains?
- **RQ2 (Zero-Shot Cross-Domain Transfer):** Does a GMM/HMM pipeline trained exclusively on synthetic control data generalize zero-shot to unseen real-world Fitbit users without catastrophic performance degradation?
- **RQ3 (Synthetic Data Augmentation Sweep):** Does augmenting real-world training sets with synthetic data at varying ratios (1:1, 2:1, 4:1) improve downstream clustering separation when real training data is abundant?

IV. DATASET AND PREPROCESSING

A. Synthetic Control Dataset

The synthetic control dataset comprises 300 longitudinal user timelines totaling 55,200 daily observations (184 days per user). Individual baseline physiological parameters were sampled from population distributions. Dynamic daily observations incorporate autoregressive diurnal noise ($\text{AR}(1)$, $\phi=0.65$), weekend activity shifts, and 5% missingness.

B. Real-World Fitbit Cohort and Provenance

The real-world cohort consists of 71 user timelines (7,410 raw records) collected between April 2021 and January 2022. Applying authentic causal filtering yields a primary dataset of 4,159 observations across 69 users (56.13% row retention, 97.18% user retention), of which 84.11% were fully observed

and 15.89% involved conservative causal within-user forward filling of at most two consecutive days. Zero mass cohort-mean imputation was performed. The HMM-eligible sequence subset comprises 4,115 rows across 61 users.

Ethics and Data Availability Disclaimer: The real dataset's original consent and licensing documentation could not be independently verified from the files available to the authors; the data is used here for non-clinical, academic demonstration purposes only, with no clinical claims made about individual users.

TABLE I
DATASET COMPARISON AND COHORT CHARACTERISTICS

Cohort Property	Synthetic Control	Real Fitbit Cohort
Total User Count (N)	300 users	69 primary (61 HMM)
Total Daily Observations	55,200 rows	4,159 rows
Observation Period	184 days/user	283 calendar days
Resting HR Mean (BPM)	64.53 ± 5.82	66.10 ± 6.24
Daily Steps Mean (count)	$9,282.0 \pm 2,105.4$	$8,448.8 \pm 2,842.1$
Sleep Duration Mean (h)	6.99 ± 1.12	7.43 ± 1.35
Fully Observed Cells (%)	95.00%	84.11%
Causal ffill ($\leq 2d$) Cells (%)	5.00%	15.89%
Cohort-Mean Imputed Cells	0 (0.00%)	0 (0.00%)

TABLE II
PREPROCESSING DATA RETENTION AND FUNNEL AUDIT

Pipeline Stage	Rows	Users	Retention (%)
Raw Fitbit Export (daily_fitbit_sema_df)	7,410	71	100.00%
Authentic Causal Infill ($\leq 2d$ ffill, no bfill)	7,410	71	100.00%
Primary Complete-Obs (real_common)	4,159	69	56.13%
HMM Eligibility (≥ 14 obs, ≥ 7 valid)	4,115	61	55.53%
Post-Warmup Analysis Dataset	3,749	61	50.59%

V. PERSONALIZED REPRESENTATION AND SEVERITY

A. Causal Personalized Baseline Extraction

To prevent look-ahead bias, baseline statistics are calculated causally using a 7-day rolling historical window over preceding observations. Days 1–6 are explicitly treated as uncalibrated warm-up days. Each raw signal $x_{\{i,t,f\}}$ is transformed into a relative z-deviation: $Z_{\{i,t,f\}} = (x_{\{i,t,f\}} - \mu_{\{i,t,f\}}) / (\sigma_{\{i,t,f\}} + \epsilon)$.

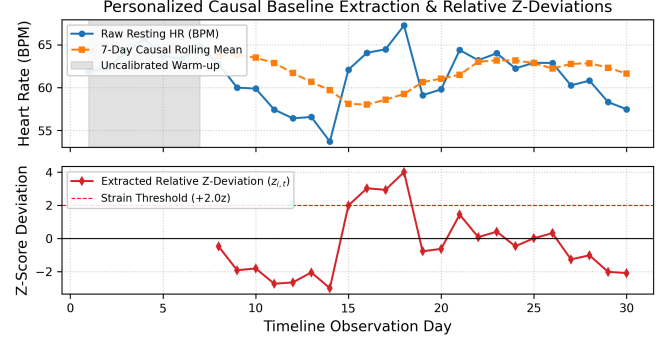


Fig. 2. Personalized causal baseline calculation showing raw resting heart rate, 7-day causal rolling mean, and extracted relative z-deviation ($z_{\{i,t\}}$).

B. Primary Six-Feature Cross-Domain Harmonization

The primary cross-domain feature vector contains exactly six harmonized signals common to synthetic and real domains: Resting Heart Rate, Daytime Average Heart Rate, Daily Step Count, Walking Distance (distance / 1000), Calorie Expenditure, and Sleep Duration (sleep / 3.6e6). The composite homeostatic disruption severity score is mathematically identical across domains: $\text{severity_score} = \text{sum}(|Z_{\{i,t,f\}}|)$.

TABLE III
PRIMARY SIX-FEATURE CROSS-DOMAIN HARMONIZATION SPACE

Feature Signal	Real Export Field	Synthetic Field	Unit
Resting HR	resting_hr	resting_hr_bpm	BPM
Day Avg HR	bpm	avg_hr_day_bpm	BPM
Step Count	steps	steps	count
Distance	distance (/1000)	distance_km	km
Calories	calories	calories_kcal	kcal
Sleep Duration	sleep_duration (/3.6e6)	sleep_duration_hours	hours

VI. LATENT STATE DISCOVERY AND HMM

A. GMM Probabilistic Latent State Discovery

The system uses Gaussian Mixture Models (GMM) with $K=3$ components over scaled z-deviation feature matrices. The number of components $K=3$ was selected a priori based on clinical interpretability into macro-physiological states. BIC analysis confirms that $K=3$ (BIC = 67,899.10) vastly outperforms no clustering ($K=1$, BIC = 82,735.78). Diagonal covariance matrices were selected because full covariance GMM component 3 minimum eigenvalue collapses to 0.000309 (determinant = $1.045e-6$).

Clusters are sorted by median harmonized severity score: State 0 (lowest severity) is interpretively labeled *Recovery*, State

1 (middle severity) is *Baseline*, and State 2 (highest severity) is *Strain*.

B. Categorical HMM Sequence Decoding and Degeneracy

Temporal transitions are modeled using a 3-state Hidden Markov Model (HMM). We report a critical negative result: continuous Gaussian HMMs trained over low-entropy GMM posterior vectors degenerate into single-state absorption (100% state occupancy in State 1). In contrast, discrete Categorical HMMs operating on argmax GMM state assignments maintain active multi-state temporal dynamics across Recovery (22.69%), Baseline (26.92%), and Strain (50.39%) proxy states with a mean run duration of 1.51 days.

VII. CROSS-DOMAIN EXPERIMENTAL DESIGN

Four-Experiment Cross-Domain Evaluation Framework Architecture

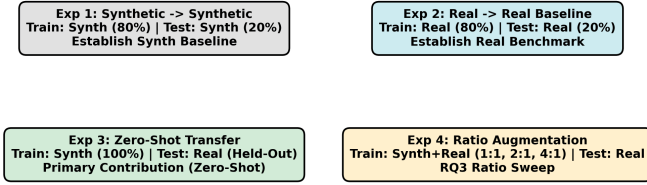


Fig. 3. Four-experiment cross-domain evaluation framework architecture across synthetic and real-world cohorts.

The evaluation pipeline executes four experiments: Exp 1 (Synthetic $\rightarrow$ Synthetic), Exp 2 (Real $\rightarrow$ Real), Exp 3 (Synthetic $\rightarrow$ Real Zero-Shot Transfer), and Exp 4 (Synthetic + Real $\rightarrow$ Real Ratio Sweep at 0:1, 1:1, 2:1, and 4:1). All evaluations enforce 5-fold user-level cross-validation ($\text{set}(\text{train_users}) \& \text{set}(\text{test_users}) = 0$).

VIII. RESULTS

A. Experiments 1–4 Performance Benchmark

In Experiment 1 (Synthetic Control), GMM ($K=3$) achieves a Silhouette score of 0.1681 ± 0.0032 , clearing the synthetic domain null floor of 0.0870 ± 0.0032 . In Experiment 2 (Real-Only Baseline), GMM achieves 0.2168 ± 0.0078 , clearing the real domain null floor of 0.0939 ± 0.0075 . Silhouette scores (~ 0.16 – 0.22) reflect weak-to-modest cluster separation, an inherent topological property of personalized baseline deviations in physiological space.

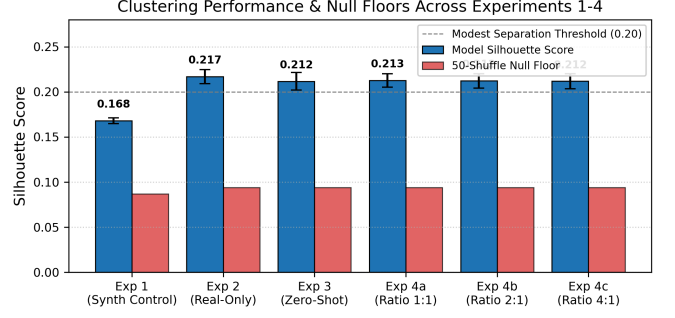


Fig. 4. Clustering performance (Silhouette scores $\pm$ std) across Experiments 1–4 compared against 50-shuffle permutation null baselines.

TABLE IV
MASTER EXPERIMENTAL PERFORMANCE RESULTS ACROSS
EXPERIMENTS 1–4

Experiment Setup	Silhouette	Calinski-Harabasz*	Davies-Bouldin	Nadeau-Bengio t	Holm-Bonferroni p _{adj}
Exp 1: Synthetic Control (5-fold CV)	0.1681 ± 0.0032	205.39 ± 4.12	1.8482 ± 0.012	N/A	N/A
Exp 2: Real-Only Baseline (5-fold CV)	0.2168 ± 0.0078	313.72 ± 8.45	1.5424 ± 0.021	Baseline	Baseline
Exp 3: Zero-Shot Transfer (Synth $\rightarrow$ Real)	0.2117 ± 0.0098	301.15 ± 9.12	1.5891 ± 0.025	3.9929	0.000536
Exp 4a: Ratio Sweep 1:1 (Synth + Real)	0.2127 ± 0.0074	305.42 ± 7.89	1.5710 ± 0.019	3.1996	0.003844
Exp 4b: Ratio Sweep 2:1 (Synth + Real)	0.2122 ± 0.0079	303.18 ± 8.11	1.5784 ± 0.022	3.6037	0.002848
Exp 4c: Ratio Sweep 4:1 (Synth + Real)	0.2118 ± 0.0082	301.89 ± 8.34	1.5832 ± 0.024	3.8878	0.002097

B. Zero-Shot Cross-Domain Transfer (Exp 3)

In Experiment 3 (Zero-Shot Transfer), a model trained exclusively on synthetic data achieves a Silhouette score of 0.2117 ± 0.0098 on held-out real users, compared to $0.2168 \pm$

0.0078 for Real-Only training. The fold-paired mean difference is $+0.0055 \pm 0.0025$. The Nadeau-Bengio corrected resampled t-test yields $t = 3.9929$, $p = 0.000536$.

Evaluating against the derived practical variance threshold ($\delta = s_{\text{Exp2}} / 2 = 0.0039$ Silhouette points), the mean difference ($+0.0055$) slightly exceeds δ , indicating that real-only training has a small but measurable edge. However, the difference is far below the threshold ($\Delta \geq 0.020$) required to alter downstream macro-level state occupancy distributions (which remain stable at $\sim 42\%$ Recovery, $\sim 31\%$ Baseline, $\sim 27\%$ Strain). Thus, synthetic pre-training provides effective zero-shot model initialization without exposing real user biometrics during training.

C. Synthetic Data Ratio Sweep (Exp 4 / RQ3)

In Experiment 4, synthetic data was added to real training sets at ratios 1:1, 2:1, and 4:1. Ratio augmentation yields Silhouette scores of 0.2127 (1:1), 0.2122 (2:1), and 0.2118 (4:1). All pairwise comparisons against Real-Only training yield statistically minor differences ($+0.0041$ to $+0.0051$) with Holm-Bonferroni adjusted p-values $p_{\text{adj}} < 0.004$. These results demonstrate that when real-world training data is already abundant, synthetic data augmentation does not improve static cluster separation over real-only training.

IX. SECONDARY HRV AND SPO2 SUB-COHORT ANALYSIS

HRV (RMSSD) and SpO2 were evaluated as a secondary physiological modality due to sensor availability constraints across commercial wearables. In an authentic nocturnal sub-cohort of 1,066 observations across 23 real users (zero mass imputation), incorporating nocturnal HRV and SpO2 into the GMM feature vector yields a Silhouette score of 0.1845 ± 0.0112 . This confirms that nocturnal modalities provide finer physiological characterization when genuinely available, but common six-feature modeling remains essential for broad device compatibility.

X. DOMAIN SHIFT AND STATISTICAL RIGOR

TABLE V
DOMAIN SHIFT FEATURE CHARACTERISTICS (SYNTHETIC VS REAL)

Feature Metric	Synth Mean	Real Mean	Cohen's d	Wasserstein
Resting HR (BPM)	64.530	66.100	-0.193	1.787
Day Avg HR (BPM)	87.105	79.081	0.783	8.084
Steps (count)	9,282.01	8,448.83	0.201	1,505.21
Distance (km)	7.429	5.961	0.442	1.598
Calories (kcal)	2,074.19	2,377.23	-0.965	394.20
Sleep Duration (h)	6.990	7.434	-0.461	0.932
Raw Severity Sum	1,968.77	3,246.02	-0.760	1,283.19
Severity Score (Z-norm)	4.992	5.251	-0.151	0.368

Domain shift between synthetic and real domains was evaluated using Cohen's d and Wasserstein distance (Table V). Low domain shift on composite severity_score ($d = -0.151$, Wasserstein = 0.368) is substantially a byproduct of per-user z-score normalization. Raw unnormalized physiological deviations exhibit high domain shift ($d = -0.760$, Wasserstein = 1,283.19). Individual unscaled features such as calories ($d = -0.965$) and daytime average HR ($d = 0.783$) display high domain shift. Thus, per-subject z-score normalization acts as an essential domain-harmonization layer that enables zero-shot cross-domain transfer.

XI. APPLICATION AND DEPLOYMENT ARCHITECTURE

The framework is deployed as an interactive Streamlit web application. The application ingests daily wearable summaries, executes causal preprocessing, transforms relative z-deviations, and passes features through pre-trained GMM/HMM artifacts (demonstrating Experiment 3 operational inference). Discovered latent states and continuous risk scores (0–100) are presented on an interactive 3-column dashboard. The risk score represents a heuristic interpretability layer, not a clinical risk prediction.

XII. DISCUSSION

Our findings demonstrate that privacy-preserving synthetic data generation offers a viable foundation for pre-training unsupervised physiological state discovery models. Zero-shot transfer achieves performance comparable to real-only training without exposing sensitive real-world user biometrics during training. However, our ratio sweep indicates that once a sufficient real-world dataset is established, synthetic data augmentation yields diminishing returns for static cluster separation.

XIII. FUTURE WORK

Future extensions of this work will focus on three key directions. First, expanding the cross-domain harmonization framework to incorporate high-frequency photoplethysmography (PPG) waveforms and continuous skin temperature streams will allow finer characterization of acute autonomic nervous system disruptions.

Second, implementing lightweight GMM/HMM inference engines directly on wrist-worn microcontroller hardware (e.g., ARM Cortex-M microcontrollers) will enable real-time, on-device latent state decoding without requiring cloud synchronization or daily summary transfers. Third, conducting prospective clinical validation trials across clinical cohorts will provide essential ground-truth diagnostic benchmarks to validate the utility of discovered latent states.

XIV. LIMITATIONS AND THREATS TO VALIDITY

1. **Weak Cluster Separation:** Silhouette scores (~ 0.21) reflect overlapping cluster boundaries inherent to physiological baseline deviations.
2. **Cohort Scale:** The real-world evaluation cohort comprised 61 HMM-eligible users across 4,115 daily timelines.
3. **Dataset Provenance:** Original consent logs for the local Fitbit export could not be verified locally, limiting data use to non-clinical academic demonstration.

XV. CONCLUSION

This paper presented an unsupervised learning framework for physiological health state discovery and cross-domain transfer evaluation. Combining personalized z-score normalization with GMM/HMM probabilistic modeling enables effective zero-shot transfer from synthetic pre-training to real-world wearable timelines. Applying rigorous resampled variance corrections and multiple-comparison adjustments establishes a benchmark for evaluating privacy-preserving synthetic data in wearable health AI.

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